

## Research papers

# A graph neural network using physical attributes to improve the system-wide nodal water-level prediction in sparsely monitored urban drainage systems

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## ABSTRACT

Nodal water levels are a critical hydraulic parameter indicative of the operational status of urban drainage networks (UDNs), and their system-wide sensing is essential for evaluating system capacity and promptly identifying urban flooding and overflow pollution risks. However, due to financial constraints and installation challenges, the widespread deployment of sensors in UDNs is impractical. While developing approaches based on graph neural networks for system-wide sensing and prediction in sparsely monitored drainage systems is an effective solution, methods that rely solely on simple topological connectivity exhibit instability and significant prediction errors due to the complex and variable flow conditions within UDNs, influenced by multiple uncertainties. To address this challenge, we propose an edge-attribute-enhanced spatiotemporal graph convolutional network (Edge-STGCN) to improve prediction accuracy in sparsely monitored UDNs, offer a novel perspective to evaluate sensor placement strategies in different branches, and analyze the predictive utility of individual and combined edge attributes for model performance. Results revealed that with only 10% of nodes monitored, the Edge-STGCN model achieved reliable predictions at 88.6% of system-wide nodes, significantly outperforming the multilayer perceptron (14.5%) and STGCN (47.0%). Pipes near the outfall, particularly small-diameter branches whose invert elevations were higher than those of the main trunk pipes, were especially prone to uncertainty in prediction accuracy. Pipe invert elevation was an important contributor to the model's prediction accuracy. The proposed method enables reliable predictions, informs sensor placement strategies, and provides data support for decision-making aimed at mitigating flooding and pollution.

## 1. Introduction

As a critical component of underground infrastructure, the urban drainage network (UDN) plays a pivotal role in city management (Wang & Yin, 2022). Proper construction and maintenance of this infrastructure are essential for effectively managing urban wastewater, stormwater, and drinking water, thus improving urban environmental quality and ensuring water safety (Yin et al., 2020). However, as the operational lifespan of these pipelines increase continuously, issues such as sediment accumulation, scaling, deformation, and leakage have become increasingly prominent (Di et al., 2023; Wang et al., 2021). Damage and sedimentation in urban drainage pipes can reduce flow capacity, causing water accumulation on the ground and inducing urban flood disasters

(Donnelly et al., 2022; Liu et al., 2024). Moreover, rapid changes in water quality and quantity can overload wastewater treatment plants, potentially leading to equipment damage, resulting in effluent quality deterioration, and economic losses (Sukias et al., 2018; Zhang et al., 2022). The intertwined distribution and concealed nature of underground pipelines pose significant challenges for the comprehensive analysis of drainage system operational loads and management risks (Gokhale & Graham, 2004). Despite the recent implementation of online monitoring equipment in smart urban drainage systems, which enables real-time monitoring of drainage network operational status by collecting key data such as water levels, flow rates, and water quality, the limited distribution of monitoring points due to financial constraints and installation challenges results in insufficient spatial and temporal

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coverage (Mao et al., 2020; Shaban et al., 2023). This limitation hinders the ability to fully understand the overall operational status of UDNs and creates blind spots that impede timely detection and response to potential issues. Consequently, achieving system-wide sensing and reliable prediction of UDNs with limited monitoring nodes to provide scientific decision support for intelligent management, has gained increasing attention from the research community.

Hydraulic models such as storm water management model (SWMM), integrated catchment management for urban drainage systems (InfoWorks ICM), and MIKE URBAN offer the capability to sense and predict the operational state of UDNs by simulating unsteady water flow, runoff, and system performance under various conditions through detailed physical models (Elliott & Trowsdale, 2007), and have been widely applied for quantitatively assessing drainage system performance, identifying flood risks, and exploring the impacts of climate change (Chen & Huang, 2024; Maryam et al., 2021; Tuomela et al., 2019). However, establishing hydraulic models requires both the structure and geometric properties of drainage networks, alongside numerous specific parameters (e.g., pipe roughness, sedimentation conditions, and high-resolution underlying surface data) for reliable performance (Ma et al., 2022; Zhang et al., 2023). Parameter acquisition requires extensive field investigation and measurement that are time-consuming, costly, and often impractical. Additionally, infrastructure aging causes dynamic changes in parameters such as pipe roughness and sedimentation levels, complicating calibration and necessitating regular recalibration (Shahed Behrouz et al., 2020). The intensive computation required for solving partial differential equations in hydraulic models poses major challenges for real-time monitoring and effective management of drainage networks, especially in resource-limited settings (Li et al., 2024).

With the development of machine learning algorithms, researchers have proposed hybrid models based on hydraulic models and machine learning algorithms to solve these challenges. The essence of hybrid models is to serve as surrogate models for hydraulic models, enabling real-time sensing and prediction of the drainage system's status (Garzón et al., 2024; Zhang et al., 2024). These models use physically meaningful synthetic data, such as water levels, flow rates, and velocities, generated by hydraulic models as the training set for machine learning algorithms (Liao et al., 2023). Although the real-time issue of hydraulic models in practical applications has been addressed by hybrid models, the dependence on simulation data means that the challenges associated with model construction and calibration remain unresolved. Consequently, the development of physically meaningful drainage system perception models using machine learning algorithms without relying on hydraulic models has become a key research focus.

Conventional machine learning algorithms in existing studies typically transform input features into vectorized forms and establish mappings to target variables by fitting hidden neurons (Ma et al., 2024; Sumer et al., 2007), without considering the spatial correlations among input features. However, in UDNs with complex and irregular topologies, the hydraulic state of each node is often influenced by pipe connectivity and flow path dynamics (Fu et al., 2020; Zhang et al., 2024). Therefore, modeling approaches that preserve and leverage such structural features are essential. This trend has spurred increasing interest in graph-based learning paradigms where a graph is defined as a general-purpose mathematical structure composed of nodes and edges that represent pairwise relationships among objects (Cheng et al., 2024).

Graph Neural Networks (GNNs) model data based on graph structures (Kipf & Welling, 2017). In contrast to conventional machine learning approaches that primarily utilize node-level features in isolation, GNNs integrate both node attributes and topological structure by encoding spatial connectivity into the graph representation (Wu et al., 2021). Leveraging the message-passing mechanism of GNNs, nodes iteratively aggregate information from their neighbors during representation learning, enabling the capture of topology-induced spatial dependencies and improving the model's capacity to characterize

complex networked systems (Deng et al., 2024). This allows GNNs with a distinctive advantage in modeling the propagation of hydraulic information through the complex network topology of UDNs. GNNs have been widely applied in the domain of water distribution networks (WDNs). Existing literature has contributed to water demand forecasting (Wang et al., 2023), state estimation (Xing & Sela, 2022), pressure reconstruction (Zhou et al., 2020), hydraulic metamodelling (Zanfei et al., 2023), and water quality prediction (Li et al., 2024) in WDNs.

However, despite their proven success in WDNs, conventional GNNs models, when directly applied to UDNs, may face challenges due to the distinct hydraulic complexities of drainage systems. Compared with pressure-driven WDNs, gravity-driven UDNs typically operate under partially full or free-surface flow conditions, exhibiting complex and variable flow regimes that are significantly influenced by factors such as rainfall, catchment conditions, pipe dimensions, and the cumulative effect of upstream inflows (Martins et al., 2018; Peleg et al., 2017). For example, in a drainage network, the water level at a particular node is not only influenced by local conditions but also by upstream and downstream nodes connected through the network, and the physical characteristics of the connecting pipes determine the changing pattern of hydraulic behavior at the node (Tran et al., 2024). Recent studies have explored the integration of physical mechanisms into GNNs to improve prediction performance in drainage networks. Zhang et al. (2024) made the first attempt to incorporate two physics-informed mechanisms, including the flow relationship between manholes and connected pipes and a flooding determination technique, into a GNN to build surrogate models in the field of drainage. As a result, GNN models achieved high flood prediction accuracy while reducing the computation time to only 10 % of that required by the hydraulic model. Garzón et al. (2024) applied an autoregressive graph neural network incorporating temporal, physical, and topological guidance mechanisms to predict nodal water depths in stormwater systems. The results demonstrated that the model achieved high overall prediction accuracy and exhibited transferability in different drainage networks. He et al. (2025) attempted to employ spatiotemporal graph convolutional networks (STGCN) for real-time reconstruction of water levels at all nodes within sparsely monitored UDNs. Results indicated that this approach could accurately reconstruct water levels at unmonitored nodes by relying solely on low-density monitoring nodes and basic network topology information (the flow relationship between manholes and connected pipes).

While GCN has been applied to UDNs, several challenges remain that limit the predictive accuracy of this method at unmonitored nodes in sparsely monitored drainage systems. (1) They rely on system-wide water level and flow data, which are impractical to obtain in real-world drainage systems (Kerimov et al., 2024; Zhang et al., 2024). (2) They focus on reconstructing current water levels rather than predicting future conditions at unmonitored nodes (He et al., 2025). (3) They often treat edge information as structural and rarely investigate the predictive value of edge attributes (e.g., pipe diameter, length, invert elevation) in model learning (Garzón et al., 2024). (4) The impact of sensor placement configurations on prediction performance in sparsely monitored systems remains unexplored, despite its practical significance for real-world deployments.

To address these problems, we propose an edge-enhanced spatio-temporal graph convolutional network (Edge-STGCN), which integrates physical attributes into the graph structure. Since edge attributes are static properties and serve as critical parameters in hydraulic models for calculating water levels and flow rates, this study adopts them as key physical features integrated into the graph structure. This study focused on: (1) proposing a modeling framework that incorporates static edge attributes into a GCN for predicting future system-wide nodal water-level within sparsely monitored UDNs, (2) comparing the performance of Multilayer Perceptron (MLP), STGCN, and Edge-STGCN in predicting water levels at system-wide nodes using data from limited monitoring nodes, (3) evaluating prediction difficulty at unmonitored nodes in different branches and investigating the impact of sensor placement on

predictive performance under sparse monitoring conditions, and (4) analyzing the internal mechanisms of the Edge-STGCN model by assessing the contributions of edge features to prediction accuracy at unmonitored nodes.

## 2. Methodology

An Edge-STGCN model is proposed and trained for system-wide prediction of UDNs with limited monitoring nodes. Fig. 1 shows the workflow of the Edge-STGCN model for predicting water levels at system-wide nodes based on limited monitoring nodes, where the details of each component are described in the subsequent sections. Section 2.1 introduces data preprocessing. Section 2.2 provides an overview of GNNs. Section 2.3 provides the structure of the Edge-STGCN model and elaborates on the main functional modules. The comparison model structure MLP and STGCN of Edge-STGCN are given in Sections 2.4 and 2.5. Section 2.6 presents the calculation methods for interpretability analysis of the Edge-STGCN model. Section 2.7 provides the evaluation metrics of training loss and prediction performance. To facilitate readers' understanding of the proposed method, we also provide a simple working example in the supplementary materials (Part F).

### 2.1. Data preprocessing

The required experimental data mainly consists of two types: time-series data and spatial topology data. The time-series data includes water-level data from monitoring nodes and rainfall data, while the spatial topology data refers to the drainage system's network topology, including physical information such as pipe diameter, pipe length, and pipe invert elevation. Details of the data sources used in this study are provided in Section 3 (Case study). The preprocessing methods for these two types of data are as follows.

#### 2.1.1. Time-series data processing method

In this study, the preprocessing of time-series data involves two main types: rainfall data  $P$  and water-level data  $H$ . Both types of data are organized as time-series vectors for each node, representing the feature variation of the respective node over time as expressed in Eq. (1) and Eq. (2). Nodes without corresponding time-series data are replaced with

zero, as shown in Fig. 2.

$$P_k = [P_{k1}, P_{k2}, \dots, P_{kt}] \quad (1)$$

$$H_k = [H_{k1}, H_{k2}, \dots, H_{kt}] \quad (2)$$

where  $P_{ki}$  represents the  $i$ -th feature value (rainfall data) of the node  $k$  in chronological order,  $H_{ki}$  represents the  $i$ -th feature value (water-level data) of the node  $k$  in chronological order.

The time-series vector of all nodes in the drainage network system can be expressed in matrix form  $P \in R^{n \times t}$  and  $H \in R^{n \times t}$ , as shown in Eq. (3) and Eq.(4). The node's time characteristic matrix  $X \in R^{n \times t \times 2}$  is formed by concatenating the two matrices along the depth channel, as expressed in Eq.(5).

$$P = [P_1, P_2, \dots, P_n] = \begin{bmatrix} P_{11} & \dots & P_{1t} \\ \vdots & \ddots & \vdots \\ P_{n1} & \dots & P_{nt} \end{bmatrix} \quad (3)$$

$$H = [H_1, H_2, \dots, H_n] = \begin{bmatrix} H_{11} & \dots & H_{1t} \\ \vdots & \ddots & \vdots \\ H_{n1} & \dots & H_{nt} \end{bmatrix} \quad (4)$$

$$X = \text{concat}(H, P) \quad (5)$$

Further clarification of the input rainfall data in the model is required. Rainfall intensity data, expressed in mm/h, represent the rainfall depth per unit time at each time step. In this study, the rainfall intensity is uniform across the entire drainage network system. To enhance modeling accuracy, the rainfall input at each node is calculated by multiplying the uniform rainfall intensity by the corresponding sub-catchment area. The rainfall volume  $P_k$  for node  $k$  at time  $t$  can be computed using Eq. (6):

$$P_k = I_t \times S_k \quad (6)$$

where  $I_t$  represents the rainfall intensity at time  $t$ ,  $S_k$  represents the sub-catchment area of node  $k$ .

#### 2.1.2. Spatial topology data processing method

In this study, the preprocessing of spatial topology data involves two main types: the drainage system's network topology and the physical

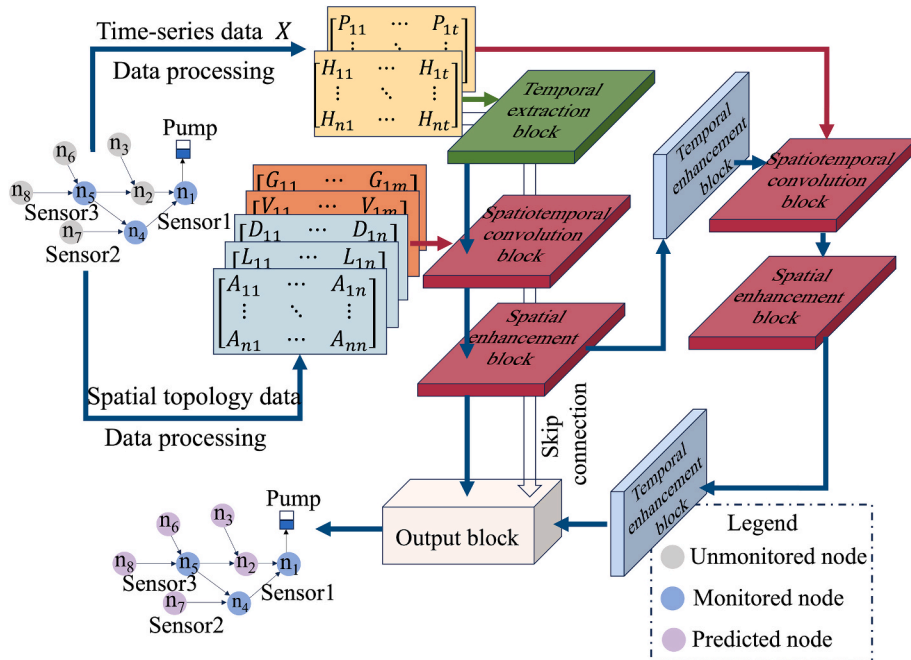


Fig. 1. The workflow of the Edge-STGCN model for predicting water levels at unmonitored nodes based on sparse monitored nodes.

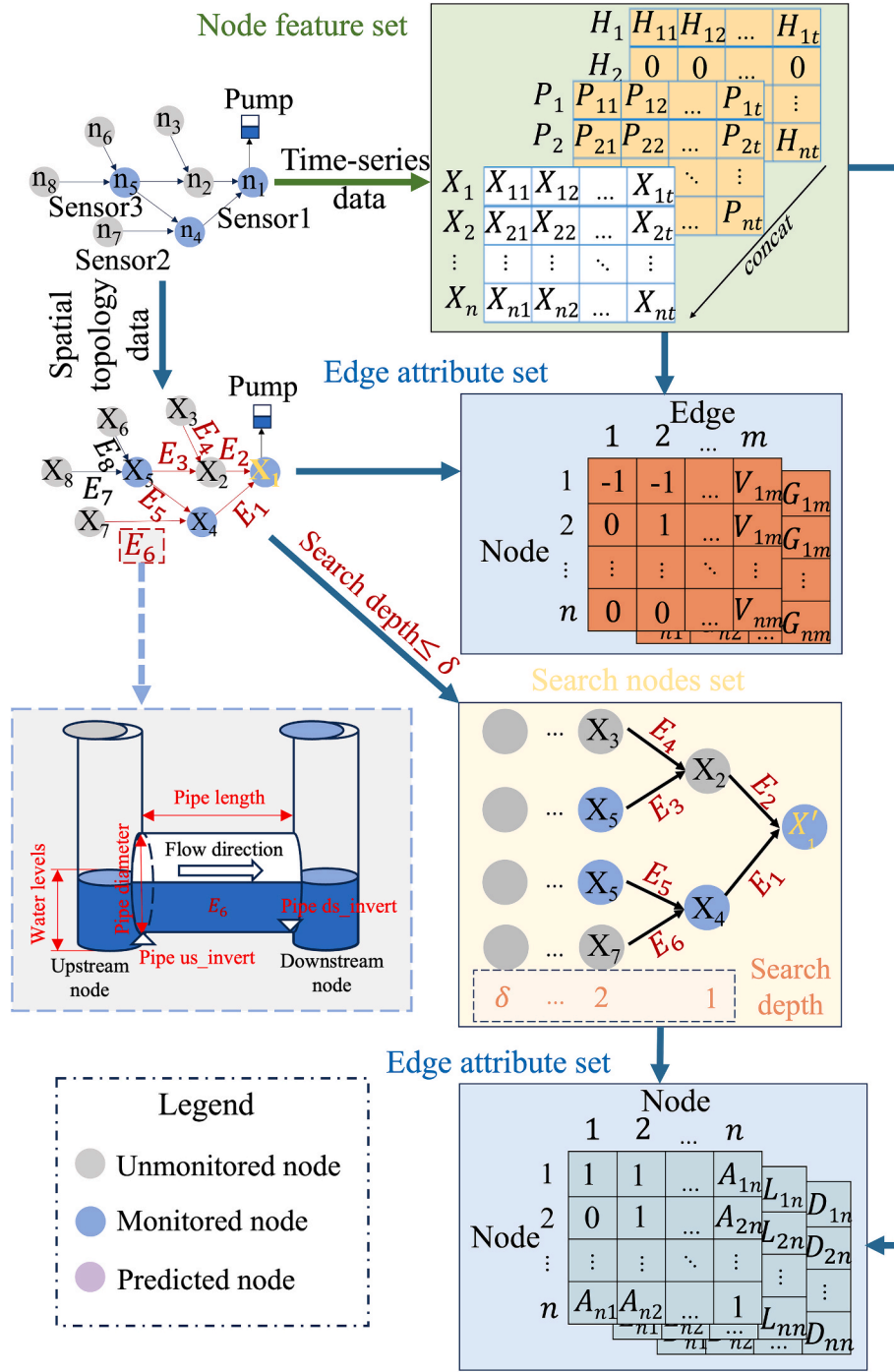


Fig. 2. The processing of time-series data and spatial topology data.

properties of the connecting pipe, including pipe diameter, pipe length, and pipe invert elevation. A node-edge mapping matrix  $V = (N, E)$  is established to process the physical information of connecting pipes, such as pipe diameter, pipe length, and pipe invert elevation, where  $N$  represents the set of nodes in the drainage network system, and  $E$  represents the set of pipes in the drainage network system (Fig. 2). The relationship between nodes and pipes is represented by matrix  $V \in R^{n \times m}$ , as expressed in Eq.(7). The invert elevation information between connecting pipes is represented by matrix  $G \in R^{n \times m}$ , as expressed in Eq.(8).

$$V = \begin{bmatrix} V_{11} & \dots & V_{1m} \\ \vdots & \ddots & \vdots \\ V_{n1} & \dots & V_{nm} \end{bmatrix} \quad (7)$$

where  $n$  represents the number of nodes and  $m$  represents the number of pipes in the drainage network system,  $V_{ij} =$

$$\begin{cases} 1, \text{node } i \text{ is the upstream node of pipe } j \\ 0, \text{node } i \text{ is not belong to pipe } j \\ -1, \text{node } i \text{ is the downstream node of pipe } j \end{cases}$$

$$G = \begin{bmatrix} G_{11} & \dots & G_{1m} \\ \vdots & \ddots & \vdots \\ G_{n1} & \dots & G_{nm} \end{bmatrix} \quad (8)$$

where  $G_{ij} = \begin{cases} g_{usij}, \text{node } i \text{ is the upstream node of pipe } j \\ 0, \text{node } i \text{ is not belong to pipe } j \\ g_{dsij}, \text{node } i \text{ is the downstream node of pipe } j \end{cases}, g_{us}$

represents the invert elevation of pipe at the upstream connecting node,  $g_{ds}$  represents the invert elevation of pipe at the downstream connecting node.

To process the connection relationships between nodes, a node–node mapping matrix  $A = (N, N)$  is established, where  $N$  represents the set of nodes in the drainage network system. The adjacency matrix  $A \in R^{n \times n}$  is expressed in Eq. (9).

$$A = \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & \ddots & \vdots \\ A_{n1} & \cdots & A_{nn} \end{bmatrix} \quad (9)$$

$$A_{ij} = \begin{cases} 1, & \text{the flow direction is from } i \text{ to } j \\ 0, & \text{the flow direction not from } i \text{ to } j \end{cases}$$

Within the specified search depth  $\delta$ , the set of nodes connected to node  $i$ , denoted as  $N_\delta$ , is identified. By applying the weight calculation formula  $f_w$ , the pipe length and pipe diameter information corresponding to the connected nodes in  $N_\delta$  are used to form the pipe length matrix  $L$  and the pipe diameter matrix  $D$ , respectively.

$$f_w(x) = \alpha e^{-\left(\frac{x-\bar{x}}{x_{std}}\right)^2} + (1-\alpha)e^{-\left(\frac{\eta}{\delta}\right)^2} \quad (10)$$

$$x_{std} = \sqrt{\frac{\sum_{i=1}^{N_\delta} (x_i - \bar{x})^2}{N_\delta - 1}} \quad (11)$$

where  $\alpha$  is the weight distribution parameter ( $\alpha = 0.65$  in this study),  $x$  is the pipe diameter or pipe length connecting node  $i$  and node  $j$  (measured in meters),  $\eta$  represents the depth level of the pipe connecting node  $i$  and node  $j$ , and  $x_{std}$  is the standard deviation of the pipe diameters in the dataset  $N_\delta$  that meets the search depth criteria.

$$L = \begin{bmatrix} L_{11} & \cdots & L_{1n} \\ \vdots & \ddots & \vdots \\ L_{n1} & \cdots & L_{nn} \end{bmatrix} \quad (12)$$

$$D = \begin{bmatrix} D_{11} & \cdots & D_{1n} \\ \vdots & \ddots & \vdots \\ D_{n1} & \cdots & D_{nn} \end{bmatrix} \quad (13)$$

where  $L_{ij} = \begin{cases} f_{w_l}(l), & \text{the length weight of pipe between nodes } i \text{ and } j \\ 0, & \text{there is no connection between nodes } i \text{ and } j \end{cases}$ ,  $l$  represents the length of the pipe connecting nodes  $i$  and  $j$  (measured in meters),  $D_{ij} = \begin{cases} f_{w_d}(d), & \text{the diameter weight of pipe between nodes } i \text{ and } j \\ 0, & \text{there is no connection between nodes } i \text{ and } j \end{cases}$ ,  $d$  represents the diameter of the pipe connecting nodes  $i$  and  $j$  (measured in meters).

## 2.2. Graph neural network

For clarity, this section provides an overview of the computational principles underlying standard GNNs. A GNN is a type of neural architecture specifically designed to operate on data structured as graphs (Kipf & Welling, 2017). In GNNs, the topological structure is represented as a graph  $G = (V, E)$ , where  $V$  is the set of nodes (e.g., manholes) and  $E$  is the set of edges (e.g., pipes). The adjacency matrix  $\bar{A}$  represents the connection relationships between nodes, as defined in Eq.(9). Firstly, each node  $v_i \in V$  is initialized as feature vectors  $h_i^{(0)}$ , which may include water levels, rainfall, or the ground elevation of each node. Subsequently, each node leverages the adjacency matrix  $\bar{A}$  to locate its neighbors and aggregates their information through message-passing mechanisms such as mean, sum, max pooling, or attention-based methods. Finally, each node updates its representation by combining its own features with the aggregated messages from its neighbors. Readers interested in a comprehensive understanding of GNNs may refer

to papers in the field (Li et al., 2023; Zhou et al., 2020). A typical update function is using following equation:

$$h_i^{(l)} = \sigma(W^l \bullet \text{aggregate}\left(\left\{h_i^{(l-1)} : j \in \mathcal{N}(i)\right\} \cup \left\{h_i^{(l-1)}\right\}\right)) \quad (14)$$

where  $\sigma$  is an activation function (e.g., ReLU),  $W^l$  is a trainable weight matrix at layer  $l$ , with  $l \in L$  and  $L$  denoting the number of neighborhood hops considered in the model (e.g., from 2-hop, 3-hop neighborhoods), and  $\text{aggregate}(\bullet)$  denotes the message-passing mechanisms.

## 2.3. Edge-STGCN model architecture

An Edge-STGCN model is proposed for system-wide prediction of drainage systems with limited monitoring nodes. The model has an  $s$ -step state input for limited monitoring nodes, spatial topology information as input, and an  $s$ -step state output of all nodes, where the input state of a node represents monitoring data, such as rainfall data and water-level data, and the output state of a node represents predicted water-level data. The Edge-STGCN model is comprised of four functional modules: temporal extraction block, spatiotemporal convolution block, spatial enhancement block, and temporal enhancement block, as shown in Fig. 1. The following sections elaborate on these functional modules.

### 2.3.1. Temporal extraction block

The temporal feature extraction block is designed to capture local dependencies in time-series data. This is accomplished through a one-layer 1D CNN that applies sliding convolutional kernels along the temporal axis to extract dynamic features within local time windows (Chen et al., 2022). The calculation formula is shown below.

$$X = \text{ReLU}(\text{CNN}(W, f(x))) \quad (15)$$

where  $X$  represents the temporal feature matrix after the convolution operation,  $x$  represents the input time-series data,  $f(\bullet)$  represents one-layer fully connected layer neural network used to map the input time-series data,  $W$  is the weight of the convolution kernel,  $\text{ReLU}$  represent rectified linear unit,  $\text{CNN}(\bullet)$  represents the convolution operation.

### 2.3.2. Spatiotemporal convolution block

The spatiotemporal convolution block is designed to expand the propagation of node temporal features in spatial dimensions, enhancing the model's ability to capture complex relationships within the drainage system. This module utilizes a two-layer GNN to propagate node temporal features across the network while introducing an optimized node–node adjacency matrix and node–edge matrix to guide the feature propagation process, in contrast to conventional GNNs with node–node adjacency matrices for feature propagation. The optimized node–node adjacency matrix and node–edge matrix incorporate pipe-specific attributes such as pipe diameter, pipe length, and pipe invert elevation, allowing the model to consider not only the direct spatial connections between nodes but also the physical characteristics of the connected pipes, which help capture the underlying hydraulic conditions. Notably, the temporal features of nodes are propagated by GNNs guided by the propagation mechanisms of pipe length, pipe diameter, and pipe invert elevation, to form the corresponding spatiotemporal feature matrix, which serves as the input of the following spatial enhancement block. A detailed explanation of the spatial enhancement block is provided in subsequent sections. The formulas for propagation aggregation of node temporal features in GNN are shown in Eq. (16)–Eq. (18).

$$X'_D = \text{GNN}((W_d^* D + b_d) \odot f(X)), A \quad (16)$$

$$X'_L = \text{GNN}((W_l^* L + b_l) \odot f(X)), A \quad (17)$$

$$X'_H = \text{GNN}((W_h^* H + b_h) \odot f(X)), V \quad (18)$$

where  $X$  represents the temporal feature matrix,  $f(\bullet)$  represents one-

layer fully connected layer neural network used to map the temporal feature matrix,  $W_d$ ,  $W_l$ , and  $W_g$ , as well as  $b_l$ ,  $b_d$  and  $b_g$  are the weights and biases of the GNNs module,  $A$  represents the adjacency matrix,  $V$  represents the node-edge mapping matrix that defines the relationship between nodes and pipes,  $*$  represents the element-wise multiplication (Hadamard product),  $\odot$  represents matrix product.

### 2.3.3. Spatial enhancement block

The spatial enhancement block is designed to capture and refine the spatial relationships between different nodes in the drainage system. This is accomplished through a two-layer 1D CNN, which extracts spatial and physical relationships from the edge attributes between nodes. The edge attributes consist of pipe diameter, pipe length, and pipe invert elevation. By integrating these edge attributes into the spatial feature extraction process, the model is able to learn how the physical properties of the pipes influence the flow dynamics between nodes.

The 1D CNN processes the edge attributes as sequential data, allowing the network to identify patterns and dependencies that exist between nodes connected by pipes. The first convolutional layer extracts basic spatial relationships, while the second layer refines these features, capturing more complex interactions, as expressed in Eq.(19) and Eq. (20). This multi-layer approach ensures that the spatial relationships between nodes, such as how water flow is affected by pipe characteristics, are thoroughly represented in the model.

$$f_i = \text{ReLU}(W_2 * \text{CNN}(\text{ReLU}(W_1 * \text{CNN}(X_i) + b_1)) + b_2) \quad (19)$$

$$F = \text{concat}(f_1, f_1, \dots, f_m) \quad (20)$$

where  $X_i$  represents the spatiotemporal feature matrix of edge attributes  $i$  after spatiotemporal convolution module operation,  $f_i$  represents the spatiotemporal feature matrix of edge attributes  $i$  after feature extraction by the spatial enhancement block,  $W_1$  and  $W_2$ , as well as  $b_1$  and  $b_2$  are the weights and biases of the CNN layer,  $m$  represents the total number of edge attributes,  $*$  represents the element-wise multiplication (Hadamard product).

### 2.3.4. Temporal enhancement block

The temporal enhancement block is responsible for capturing the hydraulic dynamics over time at each node in the drainage system. This is accomplished through a two-layer gated recurrent unit (GRU), which is well-suited for handling time-series data due to the ability to selectively remember or forget information over time (Gao et al., 2020). After the spatial enhancement block aggregates the spatial relationships between nodes, the node features enriched with physical connection information serve as input to the temporal enhancement block. The block extracts the temporal trends at each node, providing the model with an understanding of how hydraulic states evolve over time in response to spatial and temporal factors. The use of a two-layer GRU is beneficial to improve these trends, ensuring that the model captures both immediate and gradual changes in hydraulic conditions. GRU has two main gates: the reset gate and the update gate, which control the flow of information, enabling the model to selectively retain or discard information at each time step (Chung et al., 2014; Zhang et al., 2018). The formulas are expressed in Eq.(21)-Eq.(24).

$$r_t = \sigma(W_r \odot \text{concat}(\mu_{t-1}, x_t)) \quad (21)$$

$$z_t = \sigma(W_z \odot \text{concat}(\mu_{t-1}, x_t)) \quad (22)$$

$$\tilde{\mu}_t = \tanh(W \odot \text{concat}(r_t * \mu_{t-1}, x_t)) \quad (23)$$

$$\mu_t = (1 - z_t) * \mu_{t-1} + z_t * \tilde{\mu}_t \quad (24)$$

where  $x_t$  represents the current input temporal feature of node at time step  $t$ ,  $r_t$  represents the output of the reset gate,  $z_t$  represents the output of the update gate,  $h_t$  the current hidden state,  $\mu_{t-1}$  represents the

previous hidden state,  $\tilde{\mu}_t$  represents the candidate hidden state,  $\tanh$  represents the hyperbolic tangent activation function,  $*$  represents the element-wise multiplication (Hadamard product),  $\odot$  represents matrix product.

### 2.4. MLP model architecture

Owing to its simplicity and ease of implementation, the MLP is commonly adopted as a baseline model in data-driven research and has been widely applied in tasks such as flood forecasting (Adnan et al., 2023), water quality prediction (Zheng et al., 2024), and flow rate prediction (Zhang et al., 2018). As a typical structure-agnostic model, the MLP operates as a black box that treats all input features as independent and identically distributed (Merele et al., 2016). The model lacks spatial convolution and spatial feature enhancement blocks compared to Edge-STGCN model. Its input consists solely of the temporal features of each node, without considering the topological structure of the drainage network. These temporal features are first processed through fully connected layers to establish a feature mapping, then passed sequentially through a temporal feature extraction module and a temporal feature enhancement module. Finally, the processed features are fed into the output module to generate predictions. This model serves as a benchmark to evaluate the added value of incorporating graph-based spatial information.

### 2.5. STGCN model architecture

The STGCN model proposed by (He et al., 2025) comprises a temporal feature extraction block, a spatiotemporal convolution block, and a temporal enhancement block. In this study, an extended version, Edge-STGCN, is developed to explore the effects of incorporating edge attributes on system-wide water levels prediction in sparsely monitored drainage networks. Compared to the original STGCN, Edge-STGCN integrates a spatial enhancement block that embeds key physical attributes into the graph structure, thereby improving the model's ability to capture spatial flow dynamics.

### 2.6. Interpretability analysis of the model

Shapley Additive Explanations (SHAP) analysis is a method rooted in game theory, originally proposed in the field of economics to measure the contribution of participants to overall gains (Chu et al., 2024). In deep learning, it is used to calculate the contribution of features to the model's predictions, making the prediction results more interpretable (Lundberg and Lee, n.d; Wang & Peng, 2024). The calculation formula is shown below.

$$\varphi_i = \sum_{S \subseteq G \setminus \{i\}} \frac{|S|!(|G| - |S| - 1)!}{|G|!} [f(S \cup \{i\}) - f(S)] \quad (25)$$

where  $\varphi_i$  is the SHAP value for feature  $i$ ,  $G$  is the set of all features,  $S$  is a subset of features not including feature  $i$ ,  $|S|$  is the size of the subset  $S$ ,  $|G|$  is the size of the subset  $G$ ,  $f(S)$  is the model's prediction value of subset  $S$  when using only the features in  $S$ ,  $f(S \cup \{i\})$  is the model's prediction value of subset  $S \cup \{i\}$  when using the features in  $S$  and  $\{i\}$ .

### 2.7. Evaluation metrics

#### 2.7.1. Training loss

The mean-squared error (MSE) is employed as the criterion for the training loss, which quantifies the average squared deviation between the true and predicted values of a specified node. The training loss is calculated using Eq. (26):

$$\text{Loss} = \frac{1}{N_q} \sum_{i=1}^{N_q} (H_i - \hat{H}_i)^2 \quad (26)$$

where  $N_q$  is the number of sensors selected for the loss calculation,  $H_i$  is the true water-level value,  $\hat{H}_i$  is the predicted water-level value.

### 2.7.2. Prediction performance

In this study, the Nash-Sutcliffe Efficiency (NSE) coefficient is used to evaluate the level of fit between the predicted and true values over the prediction period, while mean absolute error (MAE), root mean square error (RMSE), and mean square error (MSE) are used to assess the errors between the predicted and true values. The NSE value ranges from  $(-\infty, 1]$ , with  $NSE = 1$  indicating a perfect fit between the predicted and true values over the prediction period. In this study, an NSE value  $\geq 0.5$  is considered acceptable in engineering applications (Guo et al., 2023; Pang et al., 2020). The calculation formulas Eq. (27)–Eq. (29) are shown below.

$$NSE = 1 - \frac{\sum_{i=1}^n (\hat{H}_i - H_i)^2}{\sum_{i=1}^n (H_i - \bar{H})^2} \quad (27)$$

$$MAE = \frac{1}{N_v} \sum_{i=1}^{N_v} |H_i - \hat{H}_i| \quad (28)$$

$$RMSE = \sqrt{\frac{1}{N_v} \sum_{i=1}^{N_v} (H_i - \hat{H}_i)^2} \quad (29)$$

where  $H_i$  is the true water-level value,  $\hat{H}_i$  is the predicted water-level value,  $\bar{H}$  is the mean of the true water-level values, and  $n$  is the number of system-wide nodes.

Additionally, it is necessary to understand the factors that influence predictive performance across different network segments, as more complex conditions arise due to hydraulic complexity, topology, or other factors. The entire network can be divided into branches based on junction nodes (Fig. 3). A prediction complexity index (PCI) of drainage network branches is introduced to quantitatively analyze the difficulty of predicting water levels in different branches of the drainage network. This concept is analogous to the financial terms cost rate and achievement rate (Chalfin et al., 1995). In this study, the cost is represented by the number of monitoring nodes required, and the achievement is reflected in the number of unmonitored nodes reliably predicted. A branch that requires fewer monitoring nodes (lower cost) while achieving a high number of reliably predicted unmonitored nodes (higher achievement) is considered to have a higher predictive potential and a lower

PCI. The PCI value ranges from  $[0, 1]$ , with larger PCI values indicating greater difficulty in predicting the water levels at unmonitored nodes within the branch. The calculation formula is shown below.

$$PCI_i = \left( \frac{C_i}{B_i} \right) + \left( 1 - \frac{(B_i - C_i - O_i)}{(B_i - C_i)} \right) \quad (30)$$

where  $B_i$  represents the total nodes num of branch  $i$ ,  $C_i$  represents the monitoring nodes num of branch  $i$ ,  $O_i$  represents the unmonitored nodes num with low prediction accuracy in branch  $i$ ,  $\left( \frac{C_i}{B_i} \right)$  represents the cost rate of branch  $i$ ,  $\frac{(B_i - C_i - O_i)}{(B_i - C_i)}$  represents the achievement rate of branch  $i$ .

## 3. Case study

### 3.1. Study area

The proposed method was applied to a drainage system located in southeastern China, covering an area of 159 ha with a total pipeline length of 11,932 m. Fig. 3 shows the network topology with other detailed network information. The system consists of 376 nodes (373 manholes, 1 pump station forebay, and 2 drainage outlets) interconnected by 392 pipelines. The pumping station, situated at the terminus of the system, has a design flow capacity of 10.72 m<sup>3</sup>/s.

### 3.2. Data preparation

The aim of this study was to leverage water-level data from limited monitoring nodes to achieve system-wide prediction of the nodal water-level data in the study area. The Edge-STGCN model was designed to take the water-level data from the monitored nodes with the previous 20-step and the rainfall data in the prediction 20-step as inputs, and the output being the predicted water levels for all nodes in the network for the prediction periods (20-step), with an interval of 5 min for each time step. Since obtaining monitoring data from every node in the real-world drainage system is practically infeasible, we used a calibrated SWMM to generate the dataset for model training and testing to validate the proposed approach. To simulate real-world conditions, a masking operation was applied to the input data, meaning that only the nodes with actual monitoring data were used, while the input for the unmonitored nodes was replaced with zero values. The rainfall data used in the experiment was generated using the Chicago rainfall pattern, with a 6:3:1 ratio for

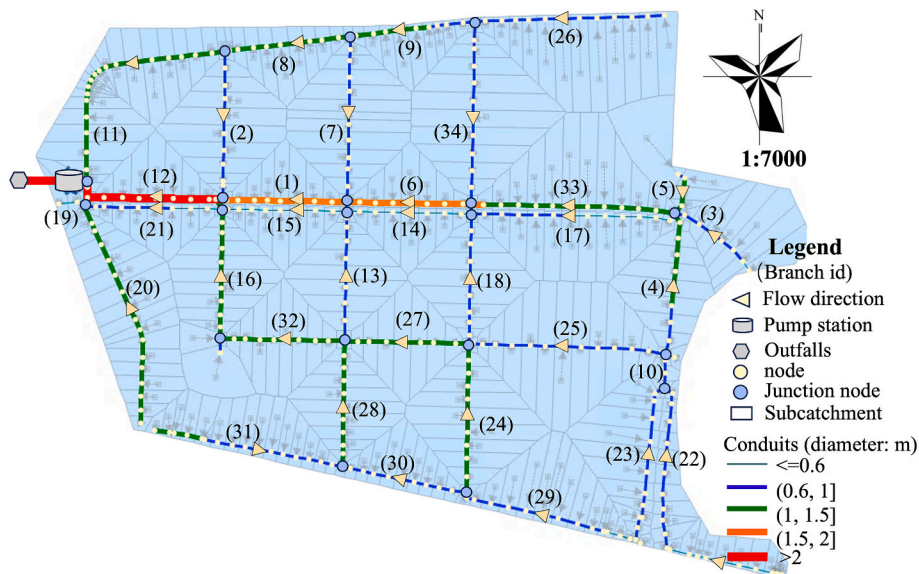


Fig. 3. The topological graph of study area.

training, validation, and testing datasets, as shown in Table 1.

### 3.3. Technical implementation

Table 2 presents the shared hyper-parameter search ranges and optimized configurations for each model of the MLP, STGCN, and Edge-STGCN, including the initial temporal feature extraction, the temporal feature enhancement, and the final output layers. Table 3 presents the hyper-parameter search ranges and optimized configurations for each model of the MLP, STGCN, and Edge-STGCN in handling the spatiotemporal feature. Model components are organized into temporal extraction, temporal enhancement, and output blocks for clarity. The optimal configurations were determined based on validation performance during training. Specifically, we employed early stopping (patience = 30) using validation MSE as the selection criterion. For each experimental condition, the model was executed in parallel 10 times, and the results were averaged. To evaluate the stability of training, all experiments were performed with 10 independent random seeds. All experiments were conducted on a GeForce RTX 4090 GPU, 16348CUDA 24 GB Turbo fan 2-slot, and AMD EPYC 7763/64Core/128Threads/2.45G/280 W/256 MB. The implementation details of the proposed models are publicly available at <https://github.com/holylove9412/Edge-STGCN-model>.

## 4. Results and discussion

### 4.1. The performance of different models in predicting system-wide nodal water levels in sparsely monitored drainage systems

With limited monitoring nodes, it is challenging for a non-graph-based deep learning model to make reliable predictions for unmonitored nodes. In graph-based models, such as STGCN, node feature information can be spatially propagated through adjacency matrices. This enables the inference of feature information for unmonitored nodes based on the similar hydraulic variation patterns of neighboring nodes. Therefore, in this section, experiments were conducted to demonstrate that STGCN could effectively utilize spatial data (connection relation between nodes) and temporal data (water levels and rainfall) in drainage network systems to explore hydraulic variation patterns between monitored and unmonitored nodes, and to assess the gain in hydraulic prediction accuracy provided by physics information, such as pipe diameter, pipe length, and pipe invert elevation.

MLP utilized temporal data, STGCN incorporated both spatial and temporal data, and Edge-STGCN integrated temporal, spatial, and edge attributes, resulting in three predictive models. These three predictive models represent: (1) a fully connected neural network that does not consider spatial dependencies, (2) a graph convolutional neural network that excludes physical edge attributes (such as pipe diameter, pipe length, and invert elevation), and (3) a graph convolutional neural network that incorporates edge attributes to guide learning, respectively. Each model was conducted 10 times in parallel under different monitoring data loss ratios, and the results were averaged. The data loss rate was defined as the proportion of nodes randomly selected from the entire network to serve as unmonitored nodes, whose water-level inputs were masked with zeros. Notably, to eliminate the impact of node distribution on model training performance, the training and validation node datasets were kept at the same locations across all models for the same data loss ratio, and the same test set was used for evaluation. The

**Table 1**

The rainfall events with different rainfall periods.

| Return period (a)                 | 1   | 2   | 3   | 5   | 10  | 15  | 20  | 30  | 50  | 100 |
|-----------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Maximum rainfall intensity (mm/h) | 110 | 131 | 147 | 166 | 193 | 208 | 220 | 235 | 255 | 282 |
| Total rainfall volume (mm)        | 48  | 60  | 67  | 76  | 88  | 96  | 101 | 108 | 117 | 129 |
| Duration (h)                      | 2   | 2   | 2   | 2   | 2   | 2   | 2   | 2   | 2   | 2   |

**Table 2**

The shared hyper-parameter search ranges and optimal configurations of the model MLP, STGCN, and Edge-STGCN.

| Module name                | Hyper-parameter        | Search range          | Optimal configuration |
|----------------------------|------------------------|-----------------------|-----------------------|
| Temporal extraction block  | Learning rate          | 0.001–0.1             | 0.001                 |
|                            | Batch size             | [16,32,64,128]        | 32                    |
|                            | Weight Decay           | 0–0.01                | 0.0005                |
|                            | Activation Function    | ReLU, Sigmoid, Tanh   | ReLU                  |
|                            | Optimize               | SGD, Adam, RMSProp    | Adam                  |
|                            | Training epoch         | [500,1000,2000,3000]  | 2000                  |
|                            | Number of CNN Layers   | [1,2,3]               | 1                     |
|                            | Number of Filters      | [16,64,128,256]       | 128                   |
|                            | Filter Size            | 3,5,7                 | 3,5                   |
|                            | Activation Function    | ReLU, Linear, Sigmoid | ReLU                  |
| Temporal enhancement block | Hidden Units           | [32,64,128]           | 64                    |
|                            | Number of GRU Layers   | [1,2,3,4,5]           | 2                     |
| Output block               | Number of Dense layers | [1,2,3]               | 2                     |
|                            | Neurons per layer      | [1,64,128,256]        | 128,1                 |
|                            | Activation Function    | ReLU, Linear, Sigmoid | Linear, ReLU          |

**Table 3**

The hyper-parameter search ranges and optimal configurations of the model MLP, STGCN, and Edge-STGCN in handling the spatiotemporal feature.

| Module name                          | hyper-parameter        | Search range          | Optimal configuration |
|--------------------------------------|------------------------|-----------------------|-----------------------|
| MLP spatiotemporal Block             | Number of Dense layers | [1,2,3]               | 2                     |
|                                      | Neurons per layer      | [64,128,256]          | 128                   |
|                                      | Activation Function    | ReLU, Linear, Sigmoid | ReLU                  |
| STGCN spatiotemporal Block           | Hidden Units           | [64,128,256]          | 128                   |
|                                      | Number of GNN Layers   | [1,2,3,4]             | 2                     |
|                                      | Activation Function    | ReLU, Linear, Sigmoid | ReLU                  |
| Edge-STGCN spatiotemporal Block      | Hidden Units           | [64,128,256]          | 128                   |
|                                      | Number of GNN Layers   | [1,2,3,4]             | 2                     |
|                                      | Activation Function    | ReLU, Linear, Sigmoid | ReLU                  |
| Edge-STGCN spatial enhancement block | Number of CNN Layers   | [1,2,3]               | 2                     |
|                                      | Number of Filters      | [16,64,128,256]       | 128                   |
|                                      | Filter Size            | 3,5,7                 | 3,5                   |
|                                      | Activation Function    | ReLU, Linear, Sigmoid | ReLU                  |
|                                      | Activation Function    | Sigmoid               | ReLU                  |

aim of this section was to provide insight into the effects of topological connections and physical information on the system-wide prediction of water levels in drainage networks with limited monitoring nodes. Therefore, during the training process of all models, monitored node data was exclusively used as inputs and for loss calculation.

Table 4 shows the average values of five indicators (MSE, RMSE,

**Table 4**

The errors between predicted and true values of system-wide nodes at monitoring data loss ratios ranging from 10% to 90% in the test set for MLP, STGCN, and Edge-STGCN model.

| Loss_Ratio | Model      | MSE          | RMSE         | MAE          | NSE_Mean     | NSE_Percent   |
|------------|------------|--------------|--------------|--------------|--------------|---------------|
| 10 %       | MLP        | 0.231        | 0.480        | 0.242        | 0.774        | 91.2 %        |
|            | STGCN      | 0.377        | 0.614        | 0.371        | 0.603        | 85.1 %        |
|            | Edge-STGCN | <b>0.055</b> | <b>0.232</b> | <b>0.128</b> | <b>0.943</b> | <b>99.0 %</b> |
| 20 %       | MLP        | 0.426        | 0.653        | 0.351        | 0.531        | 80.9 %        |
|            | STGCN      | 0.367        | 0.605        | 0.368        | 0.624        | 85.1 %        |
|            | Edge-STGCN | <b>0.054</b> | <b>0.232</b> | <b>0.133</b> | <b>0.941</b> | <b>98.9 %</b> |
| 30 %       | MLP        | 0.629        | 0.793        | 0.467        | 0.268        | 71.9 %        |
|            | STGCN      | 0.390        | 0.624        | 0.393        | 0.567        | 80.9 %        |
|            | Edge-STGCN | <b>0.080</b> | <b>0.283</b> | <b>0.149</b> | <b>0.908</b> | <b>97.5 %</b> |
| 40 %       | MLP        | 0.843        | 0.918        | 0.594        | 0.046        | 59.8 %        |
|            | STGCN      | 0.439        | 0.662        | 0.415        | 0.455        | 79.0 %        |
|            | Edge-STGCN | <b>0.089</b> | <b>0.298</b> | <b>0.160</b> | <b>0.893</b> | <b>97.3 %</b> |
| 50 %       | MLP        | 1.020        | 1.010        | 0.710        | -0.142       | 46.2 %        |
|            | STGCN      | 0.498        | 0.705        | 0.446        | 0.435        | 73.0 %        |
|            | Edge-STGCN | <b>0.109</b> | <b>0.330</b> | <b>0.181</b> | <b>0.875</b> | <b>95.7 %</b> |
| 60 %       | MLP        | 1.219        | 1.104        | 0.824        | -0.370       | 33.2 %        |
|            | STGCN      | 0.638        | 0.798        | 0.478        | 0.317        | 72.3 %        |
|            | Edge-STGCN | <b>0.108</b> | <b>0.328</b> | <b>0.187</b> | <b>0.829</b> | <b>95.3 %</b> |
| 70 %       | MLP        | 1.375        | 1.172        | 0.898        | -0.562       | 25.6 %        |
|            | STGCN      | 0.743        | 0.858        | 0.544        | 0.250        | 59.5 %        |
|            | Edge-STGCN | <b>0.128</b> | <b>0.356</b> | <b>0.210</b> | <b>0.717</b> | <b>94.2 %</b> |
| 80 %       | MLP        | 1.504        | 1.226        | 0.959        | -0.762       | 21.3 %        |
|            | STGCN      | 0.728        | 0.852        | 0.556        | 0.223        | 60.3 %        |
|            | Edge-STGCN | <b>0.137</b> | <b>0.370</b> | <b>0.223</b> | <b>0.682</b> | <b>92.3 %</b> |
| 90 %       | MLP        | 1.877        | 1.370        | 1.044        | -0.966       | 14.5 %        |
|            | STGCN      | 0.878        | 0.936        | 0.662        | 0.001        | 47.0 %        |
|            | Edge-STGCN | <b>0.217</b> | <b>0.466</b> | <b>0.277</b> | <b>0.593</b> | <b>88.6 %</b> |

MAE, NSE\_Mean, NSE\_Percent) used to assess the errors between predicted and true values of system-wide nodes at monitoring data loss ratios ranging from 10 % to 90 % in the test set for the three predictive models. Specifically, NSE\_Mean refers to the average NSE coefficient computed system-wide nodes, while NSE\_Percent denotes the percentage of nodes with an NSE coefficient greater than 0.5. Comparison of predicted and true water levels at individual nodes with different accuracy levels, as shown in Fig. S1. The results indicate that as the proportion of available monitoring data decreases, the prediction accuracy of water levels for system-wide nodes declines in all models. This is an expected outcome, as less monitoring data reduces the amount of direct information available for the models to learn from. However, the extent to which each model was affected varies significantly, and understanding these differences is crucial for identifying the most suitable model for real-world applications with limited monitoring resources.

In graph-based models, feature information was propagated based on spatial topological relationships, which provided the connections between monitored and unmonitored nodes and played a critical role in water flow dynamics, allowing for the acquisition of relevant information about unmonitored nodes from neighboring monitored nodes. In contrast, the MLP model relied entirely on the training process to infer unmonitored node behavior from monitored nodes, as it did not explicitly provide for the spatial relationships between nodes. This lack of spatial awareness became particularly detrimental when monitoring data was sparse, as there were fewer direct observations for the MLP model to learn from. As a result, it can be observed that once the monitoring data loss ratio exceeded 10 %, the prediction accuracy of water levels for all network nodes using the MLP model was significantly lower than that of the graph-based model, highlighting the limitations of

non-graph-based models in handling networked systems such as drainage networks.

When comparing the models at higher levels of monitoring data loss (e.g., 90 % of nodes unavailable), the Edge-STGCN that incorporated the edge attributes model demonstrates a notable advantage. Specifically, the percentage of system-wide nodes with an NSE coefficient greater than 0.5 in the accuracy of predicting water levels reached 88.6 % for the Edge-STGCN model, whereas this percentage dropped to 47.0 % for the STGCN model and 14.5 % for the MLP model. This performance advantage of the Edge-STGCN can be attributed to the incorporation of physical parameters, such as pipe diameter, pipe invert elevation, and pipe length, which collectively influence the speed, direction, and pressure of water flow, serving as a critical role in maintaining hydraulic balance. Additional validation experiments using real-world rainfall events, with data obtained from the local drainage company, are provided in the supplementary material (Part B), which further demonstrates the applicability of the Edge-STGCN model under practical operating conditions.

#### 4.2. The impact of different sensor layouts on the prediction accuracy of system-wide nodal water levels in sparsely monitored drainage systems

In real-world scenarios, the placement and number of sensors are concentrated in a few locations due to limitations in geographic conditions and operational costs, making it common for UDNs with incomplete system-wide data coverage. In the previous section, we demonstrated that the Edge-STGCN model could effectively leverage topological connections and edge attributes to propagate monitoring information and explore potential hydraulic behaviors, achieving comprehensive perception and reliable prediction (88.6 %) of urban drainage systems with limited monitoring nodes. Therefore, in this section, the Edge-STGCN model was utilized to investigate the influence of sensor layout on the predictive performance of unmonitored nodes in UDNs with limited monitoring nodes, aiming to provide insights for optimizing sensor resource allocation and improving system-wide prediction accuracy.

First, the model performance under different sensor configurations with 10 % of the monitoring data was compared in this section. Fig. 4 shows the spatial distribution of water-level prediction errors for all unmonitored nodes under 10 randomly generated monitoring node configurations by the Edge-STGCN model. The figure shows that due to the varying distribution of monitoring nodes, prediction accuracy differed across nodes, resulting in variations in the overall mean prediction accuracy. Despite these variations, the overall prediction accuracy remained above 81 %, demonstrating the robustness and stability of the Edge-STGCN model in scenarios with sparse monitoring nodes. However, it is important to note that some unmonitored nodes, despite being located far from the monitoring nodes, still achieve high prediction accuracy, while some unmonitored nodes that were closer to the monitoring nodes exhibited relatively lower accuracy. This observation highlights that the spatial proximity between monitored and unmonitored nodes alone does not fully account for the variation in prediction accuracy.

To further explore the factors influencing the prediction accuracy of unmonitored nodes, this study divided the entire drainage pipe network into 34 branches based on junction nodes (Fig. 5b). For branches such as 28, 30, and 22, which were located near the upstream end of the drainage network and exhibited relatively simple hydraulic behavior, predictions were effectively made using sensor data from adjacent branches, as shown in Fig. 4(e) and Fig. 4(h). Conversely, for downstream branches such as 21 and 20, where hydraulic conditions were complex and variable, the prediction accuracy remained unsatisfactory with sensors installed on these branches, as shown in Fig. 4(a) and Fig. 4(j).

Branches such as 26, 31, 22, 3, and 5 were defined as headwater branches, which were located at the uppermost reaches of the drainage

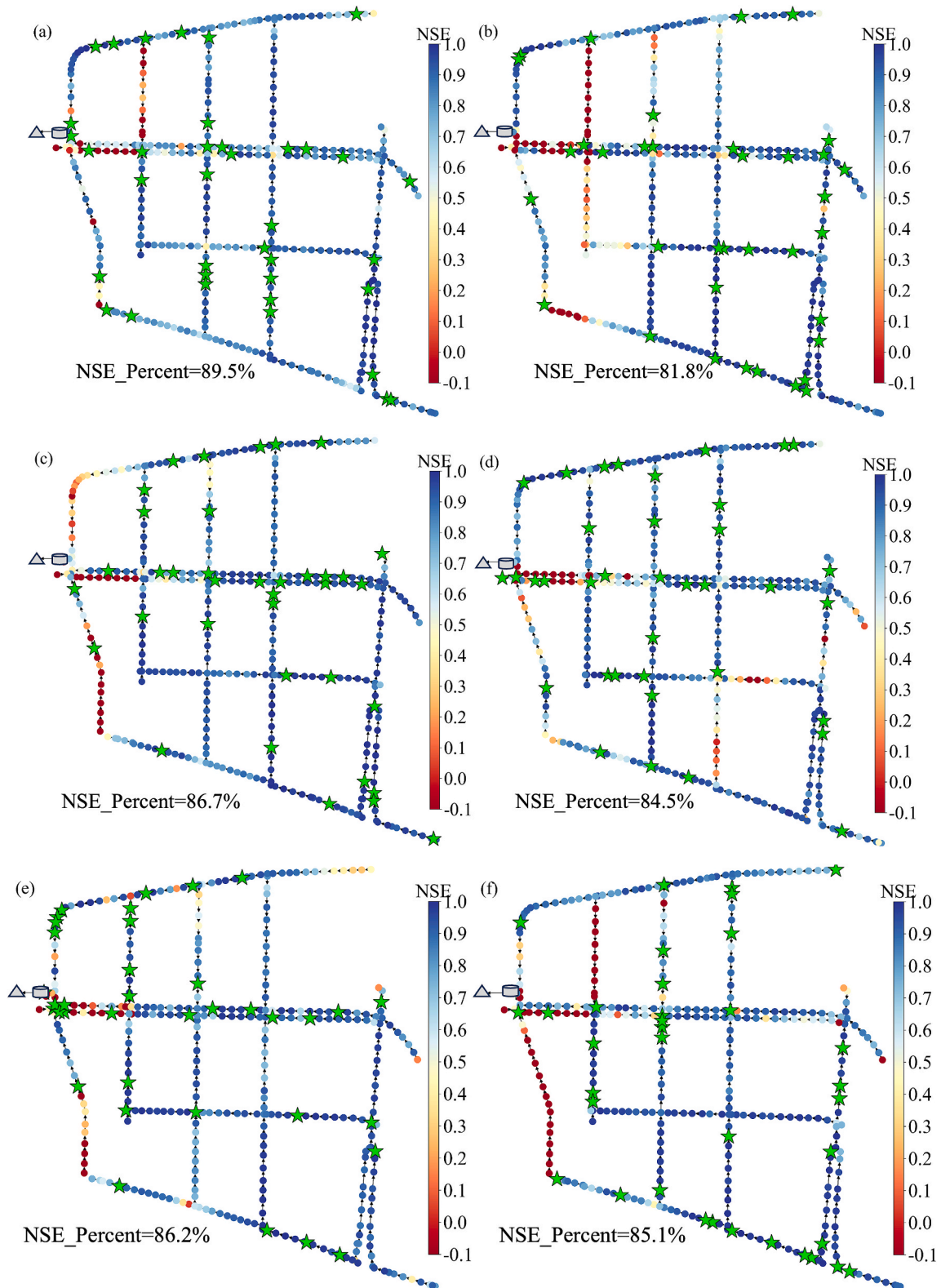


Fig. 4. The spatial distribution of system-wide node water-level prediction errors.

network and positioned far from the outfall. These branches served as the initial collection points for surface runoff and marked the starting locations of flow convergence within the system. The downstream branch connected to a headwater branch could be singular or multiple. For instance, headwater branch 22 was connected to one downstream

branch (branch 10), whereas headwater branch 26 was connected to two downstream branches (branches 34 and 9). In terms of prediction performance, all unmonitored nodes on branch 22 could be reliably predicted by deploying sensors on its downstream branch 10, as shown in Fig. 4(h). In contrast, for branch 26, sensor deployment on its

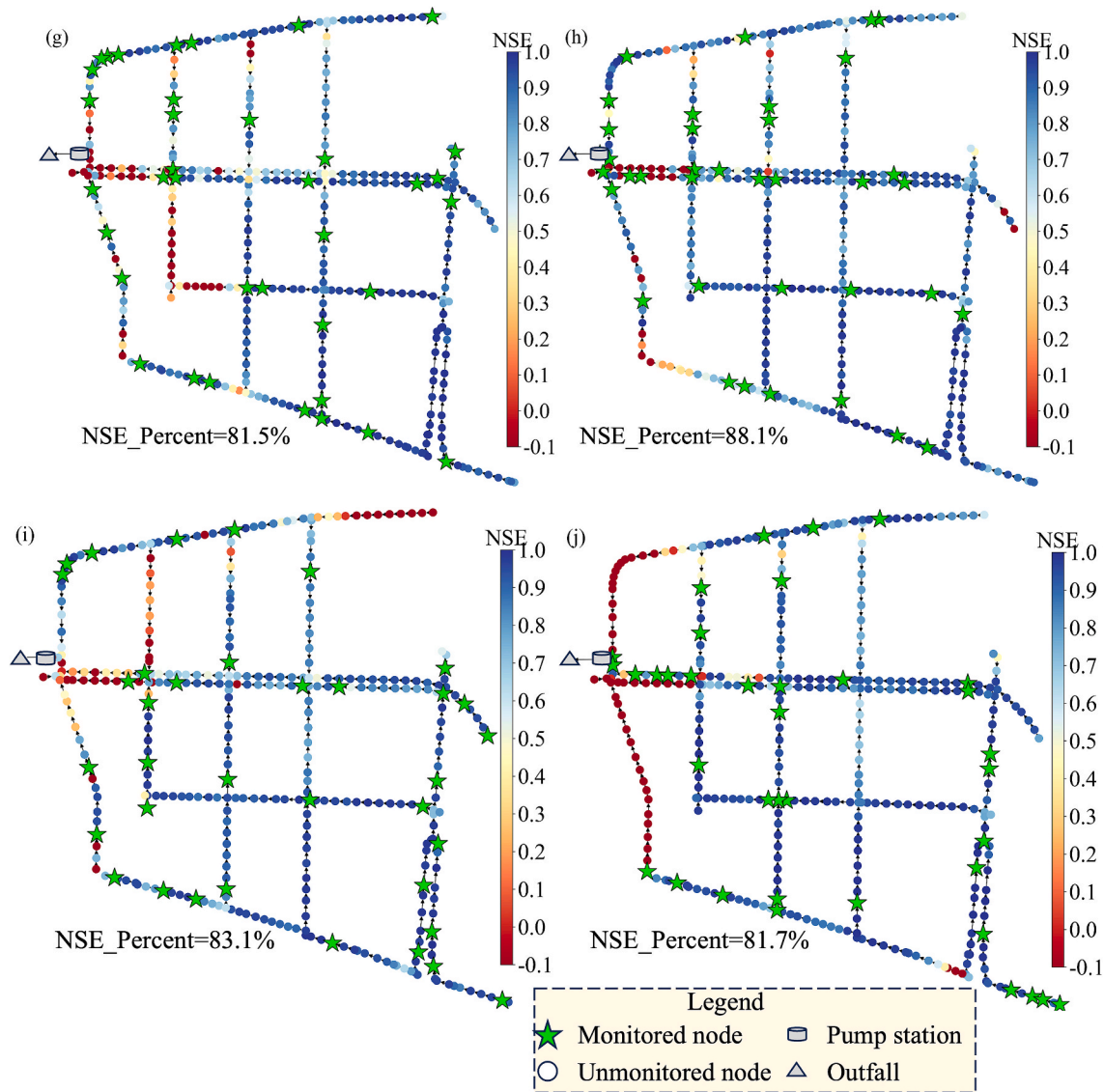


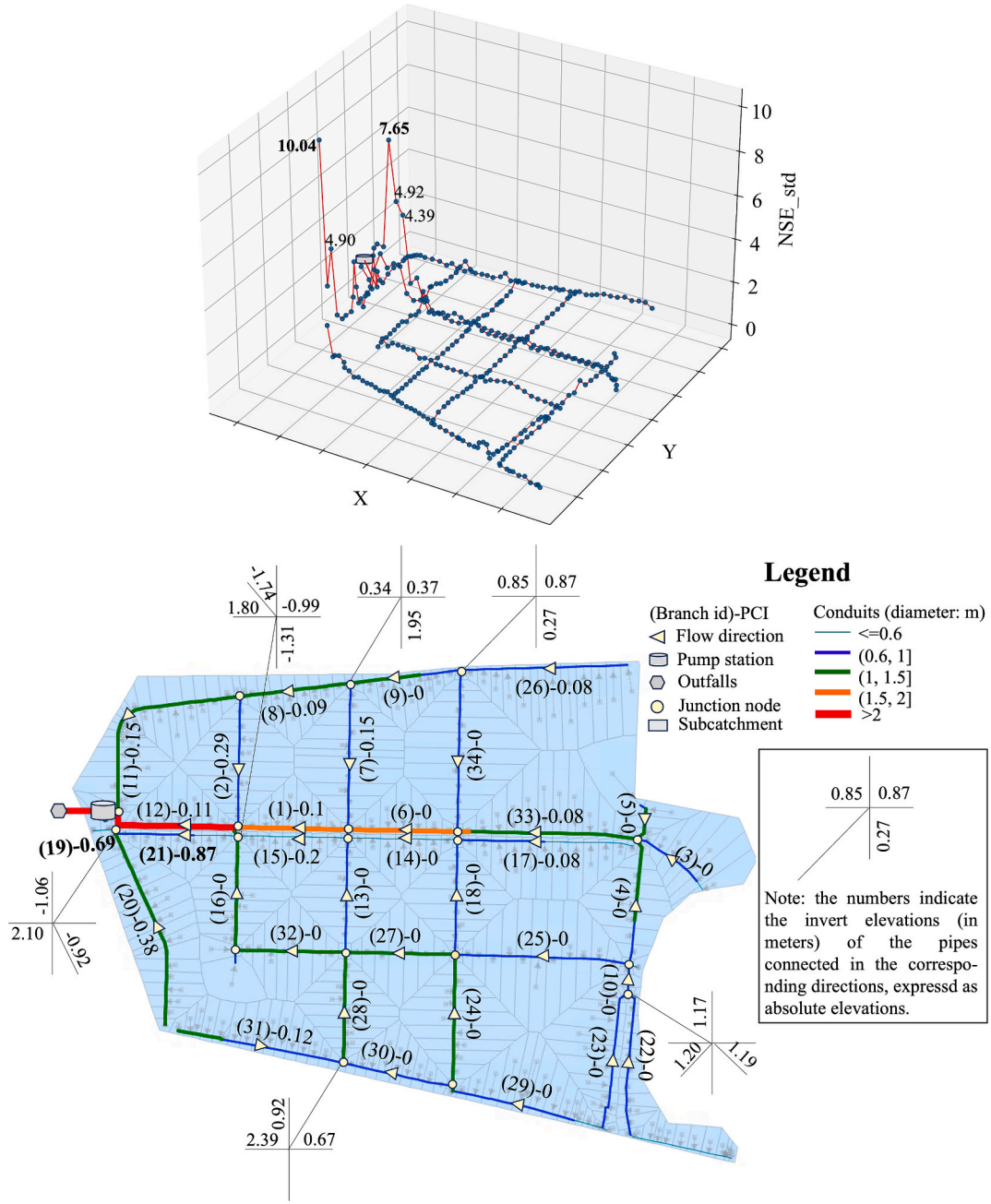
Fig. 4. (continued).

downstream branches (branches 34 and 9) alone was insufficient to achieve reliable prediction accuracy, as shown in Fig. 4(e) and Fig. 4(i). This phenomenon could be attributed to the elevation differences of connected pipes at the junction nodes. Although both branches 34 and 9 were downstream of branch 26, the invert elevation of branch 34 was substantially lower than that of branches 26 and 9, as shown in Fig. 5(b). The water level dynamics in branch 9 might have exhibited stronger hydraulic correlation with those in branch 26. When sensors were positioned on branch 9, especially at locations closed to the junction with branch 26, the prediction accuracy for unmonitored nodes on branch 26 was improved compared to placing sensors on branch 34. Details of the pipe invert elevations at junction nodes are provided in the supplementary material (Fig. S3). Therefore, this structural difference in downstream connectivity may influence flow distribution and increase the complexity of hydraulic responses in headwater branch. In general, these phenomena demonstrate the Edge-STGCN model's ability to effectively leverage monitoring node information to predict the behavior of most unmonitored nodes, predicting unmonitored nodes in certain locations remains a challenge.

Subsequently, the prediction difficulty for each branch was quantified using PCI values to enable a systematic analysis of the challenges with unmonitored node prediction in different branches. The

corresponding results are presented in Fig. 5b, where the invert elevations of pipes connected to key junction nodes are also illustrated. Invert elevations of pipes connected to the remaining junction nodes are provided in supplementary material Fig. S3.

The NSE standard deviation was calculated for each node over 10 parallel experiments to evaluate the volatility in prediction accuracy (Fig. 5a). The results indicate that unmonitored nodes with higher volatility were predominantly located near the system's end. This can be attributed to the nodes near the system's end receiving cumulative flow from multiple upstream sources, making the flow, velocity, and water level in the pipes more susceptible to upstream uncertainties, resulting in a more complex hydraulic state of nodes in these regions. The predictive potential was evaluated for each branch with limited monitoring nodes in the drainage network through the statistically analyzed results of 10 parallel experiments. For each branch, the minimum number of monitoring nodes required to achieve the highest prediction accuracy (the largest number of unmonitored nodes with NSE coefficient  $>0.5$ ) across the 10 experiments was recorded. The results are summarized in Table S1. Furthermore, the PCI was calculated for each branch to quantify the complexity of predicting water levels at unmonitored nodes based on the prediction potential for each branch, as shown in Fig. 5b. It was observed that branches 21 and 19 near the end of the system,



**Fig. 5.** The NSE standard deviation distribution of each node over 10 parallel experiments in prediction accuracy (a) and the prediction complexity index (PCI) of each branch with limited monitoring data in the drainage network by the Edge-STGCN model (b).

exhibited the highest PCI values of 0.87 and 0.69, respectively. Further analysis revealed that although the plan view showed branch 12 connected to branches 15 and 16, its invert elevation was significantly lower than those of the latter two. This elevation difference cause flows from branches 15 and 16 to preferentially enter branch 12 rather than branch 21. Only when the water level at the junction of branch 12 exceeds the invert elevation of branch 21 does the flow begin to divert through branch 21. As a result, upstream flow fluctuations can trigger abrupt water level changes in branch 21, leading to reduced prediction accuracy. Moreover, the small diameter of branch 21 limits its capacity to buffer flow variations, making it more sensitive to hydraulic disturbances and further degrading predictive performance. A similar explanation applies to the behavior observed in branch 19. These findings suggest that pipes near the outfall, particularly small-diameter branches whose invert elevations are higher than those of the main trunk pipes,

are especially prone to uncertainty in prediction performance under sparse monitoring conditions.

In summary, some branches with relatively simple hydraulic behavior, such as upstream branches of the drainage network, can rely on a smaller number of monitoring nodes placed in other branches, leveraging the Edge-STGCN model to predict water levels with reliable accuracy. However, branches with more complex hydraulic behavior, such as those near the system's outfall, may require additional physical constraints or a higher number of monitoring nodes placed closer to the unmonitored nodes for reliable predictions.

#### 4.3. Assessment of the contribution of different edge attributes to system-wide nodal water levels prediction accuracy in sparsely monitored drainage systems

In predicting unmonitored nodes in UDNs using the Edge-STGCN model, the type and number of edge attributes may affect model performance. Therefore, we conducted experiments in sparsely monitored UDNs to evaluate the performance of different edge-attribute models, including the impact of a varying number of edge attributes on prediction accuracy. By using SHAP values to quantify the contribution of edge attributes, such as pipe diameter, pipe length, and pipe invert elevation, on model performance, we could gain insights into the underlying hydraulic behavior modeled by Edge-STGCN. These insights could guide data collection strategies to optimize model performance and resource allocation, especially in real-world scenarios where acquiring comprehensive network topology data is challenging.

As shown in Fig. 6, the pipe invert elevation contributed more to the model's prediction of system-wide nodal water levels in UDNs with sparse monitoring than pipe diameter and pipe length, with pipe length contributing more than pipe diameter. This can be explained by fundamental hydraulic principles: pipe invert elevation directly affects the gravitational potential energy driving water flow, which determines both the flow velocity and direction. In drainage networks, where flow is primarily gravity-driven, differences in invert elevation create either acceleration or deceleration of water flow. Although pipe diameter and length affect flow capacity and travel time, their influence on water level prediction is localized and less critical, as these attributes primarily control local dynamics rather than the overall hydraulics, compared to pipe invert elevation. We provide the learned weights associated with the static edge features in the network as part of the supplementary material (Part E). The results are consistent with the SHAP-based feature

importance ranking, jointly supporting the critical role of invert elevation in influencing the model's predictive performance.

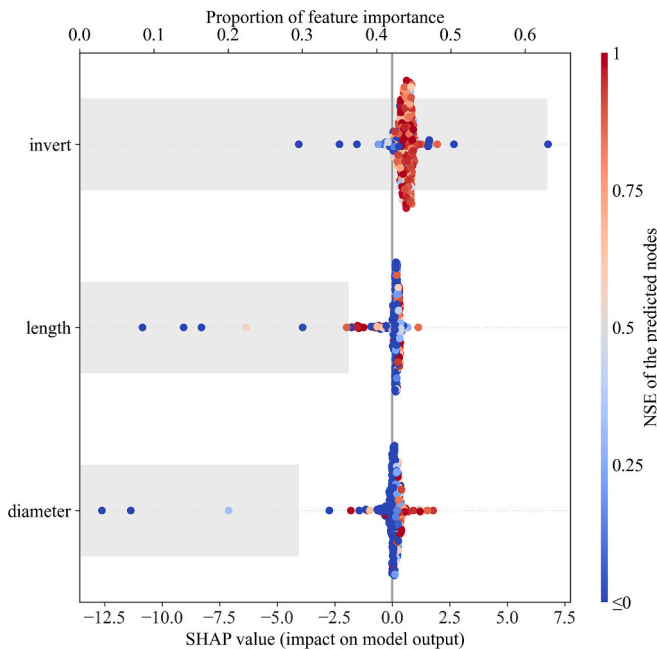
To explore the interaction effects between different edge attributes, a feature ablation experiment was conducted in which masking operations were used to evaluate the model's prediction errors under varying edge attribute conditions (Fig. 7). It can be observed that multiple features improved the model's prediction accuracy compared to a single feature. The combination of pipe diameter and pipe length as input features significantly improved the model's prediction performance compared to using them individually, equaling the prediction performance of using pipe invert elevation, with MSE values reduced by 71.9 % to 76.0 %. When pipe invert elevation was combined with either pipe length or pipe diameter, the model's prediction performance saw a slight improvement compared to using pipe invert elevation alone, with MSE values reduced by 8.4 % to 22.9 %. The best prediction performance was achieved when pipe diameter, pipe length, and pipe invert elevation information were all used as input features, with MSE values reduced by 34.7 % compared to using pipe invert elevation alone. These findings suggest that the integration of pipe diameter (flow capacity) and pipe length (travel time) into pipe invert elevation may help the model in capturing patterns of water flow dynamics, particularly in scenarios where these attributes jointly modulate flow behavior. It is important to note that these results reflect model-specific internal behavior rather than causal relationships, and may vary under different model architectures. Nevertheless, the analysis offers valuable engineering insights into which physical attributes the model predominantly relies on when predicting system-wide water levels in sparsely monitored UDNs.

From a hydraulic perspective, these findings are consistent with fundamental principles of flow dynamics and provide theoretical support for the conclusions of this study. Pipe invert elevation is the primary driver of gravitational potential, controlling the velocity and direction of water movement. However, pipe diameter determines the volume of water that can be carried through the system, and pipe length impacts the travel time from one point to another. When combined, these attributes provide a comprehensive representation of the hydraulic behavior within the network.

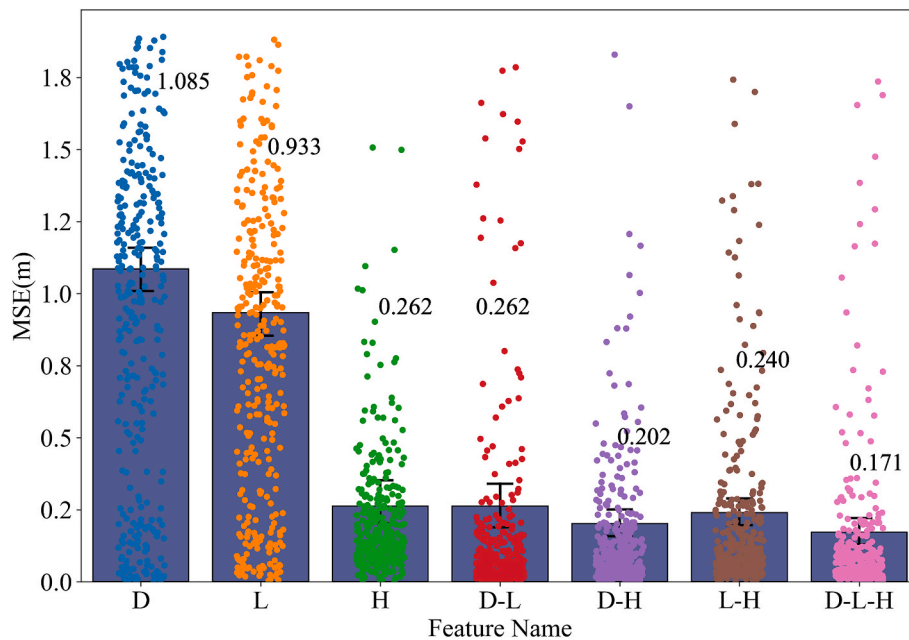
## 5. Conclusions

This study proposed an Edge-STGCN model to achieve comprehensive perception and reliable prediction in urban drainage systems with sparsely monitored nodes, offered a novel perspective to evaluate sensor placement strategies in different branches, and analyzed the predictive utility of individual and combined edge attributes for model performance. The proposed method incorporated important physical factors influencing hydraulic behavior, such as pipe diameter, pipe length, and pipe invert elevation, as guiding mechanisms to better capture water flow dynamics. The results support three major conclusions:

- (1) The Edge-STGCN model, incorporated with physical edge attributes, could effectively learn hydraulic behavior in UDNs and outperformed conventional MLP and STGCN models in predicting water levels at system-wide nodes, under sparsely monitoring data conditions.
- (2) The Edge-STGCN model could reliably predict water levels for unmonitored nodes with simple hydraulic behavior (e.g., headwater branches) using information from distant monitoring nodes. However, pipes near the outfall, particularly small-diameter branches whose invert elevations are higher than those of the main trunk pipes, are especially prone to uncertainty in prediction performance. These findings emphasize the need for additional physical constraints or increased monitoring in these critical regions.
- (3) Pipe invert elevation was the important edge attribute among those studied for enabling the Edge-STGCN model to reliably predict water levels in UDNs. Additionally, the inclusion of pipe



**Fig. 6.** The SHAP analysis diagram of the Edge-STGCN model. Note that the y-axis is sorted by the average impact of each feature on the model's predictions (i.e., the absolute mean of SHAP values), with the most important features at the top. The x-axis shows each feature's contribution to the model's prediction. Positive values indicate an increased likelihood of the predicted outcome, while negative values indicate a decreased likelihood. Each point represents a node in the drainage network, colored by the accuracy of the model's prediction (NSE), with red and blue indicating high and low values, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



**Fig. 7.** Feature ablation analysis of edge attribute combinations. Note that D, L, and H represent pipe diameter, pipe length, and pipe invert elevation, respectively. Feature combinations are denoted by concatenating the corresponding feature letters, e.g., D-L represents the combination of pipe diameter and length.

diameter and pipe length further improved overall prediction accuracy.

The improvement in prediction accuracy achieved by incorporating physical attributes into the Edge-STGCN model highlights the crucial role of domain-specific knowledge in machine learning applications. In many engineering and environmental systems, purely data-driven approaches may struggle to capture the underlying physical processes. Such approaches offer a promising direction for future research and applications in similar systems. Although the proposed model has been validated under various rainfall conditions, it has not yet been tested across multiple network topologies. This may limit the assessment of its generalizability in structurally different urban settings. In future work, we plan to extend our experiments to multiple cities and drainage systems with diverse topological configurations. Additionally, the system-wide nodal water-level prediction results can further support various intelligent technologies for the management of UDNs, such as urban flood risk forecasting and pipeline health diagnostics, and combined studies with these technologies are recommended for future research.

#### CRediT authorship contribution statement

**Li He:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jun Nan:** Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Xuesong Ye:** Writing – review & editing, Validation, Methodology, Formal analysis. **Lei Chen:** Writing – review & editing, Validation, Software, Methodology. **Shasha Ji:** Investigation, Data curation. **Zewei Chen:** Investigation, Data curation. **Qiliang Xiao:** Methodology.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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#### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2025.134306>.

#### Data availability

The data will be made available on request. The implementation details of the proposed models are publicly available at <https://github.com/holylove9412/Edge-STGCN-model>

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